English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 20 April 2024 Test 1 ——— *The exact grading formula depends on the results. As an approximation, use 2 + the number of points. Time allowed: 3 hours. Good luck!*

Problem 1. 1. (0.25 points) An urn contains 2 yellow, 3 green and 4 blue pebbles. Draw one pebble, then roll one standard die if it is yellow, two if green, three if blue. Given a sum of 3, what is the probability the pebble was green?

2. (0.25 points) Roll a standard die five times. Find the probabilities of the events A: the maximum result is at least 3; and B: the product is divisible by 10.

3. (0.25 points) Seven balls are independently placed at random in seven boxes. Let X_i count boxes with exactly i balls. Find the distribution of X_3, $\mathbb{E}[X_3]$, and $\sum_{i=1}^{7} i\mathbb{E}[X_i]$.

Problem 2. An examination contains 15 questions with five answer choices each. A student knows ten answers but, owing to nerves, answers each correctly with probability 90%. For the other five, the student guesses at random. Let X count all correct answers. Find:

1. (0.5 points) $\mathbb{P}(X > 12)$, $\mathbb{E}[X]$ and $\mathrm{Var}(X)$;

2. (0.5 points) the probability that all correct answers were to known questions, given exactly three correct answers.

Problem 3. 1. (0.25 points) Let X be a random number from $\{0, 1, 2\}$ and let $Y \sim \mathrm{Ber}(2/3)$ be independent. Find the joint distribution and correlation of X and $Z = (X + Y) \pmod 3$.

2. (0.25 points) Let X be uniform on $\{1, \ldots, 6\}$. Find its probability generating function and express $\mathbb{E}[X]$ and DX in terms of it.

3. (0.5 points) For nonnegative integer-valued X, Y, does independence imply $g_X(s)g_Y(s) = g_{X+Y}(s)$ for every $s \geq 0$? Does the converse hold? Prove your claims.

Problem 4. Let n be a positive integer. Choose u_1 uniformly from $[1, n]$, u_2 uniformly from $[1, u_1 - 1]$, and so on until $u_\ell = 1$. Let $E_n = \{u_1, \ldots, u_\ell\}$.

1. (0.5 points) Find $\mathbb{P}(k \in E_n)$ for $k \leq n$.

2. (0.25 points) Find $\mathbb{P}(2 \in E_n \mid 3 \notin E_n)$.

3. (0.5 points) Find $\mathbb{E}[|E_n|]$.